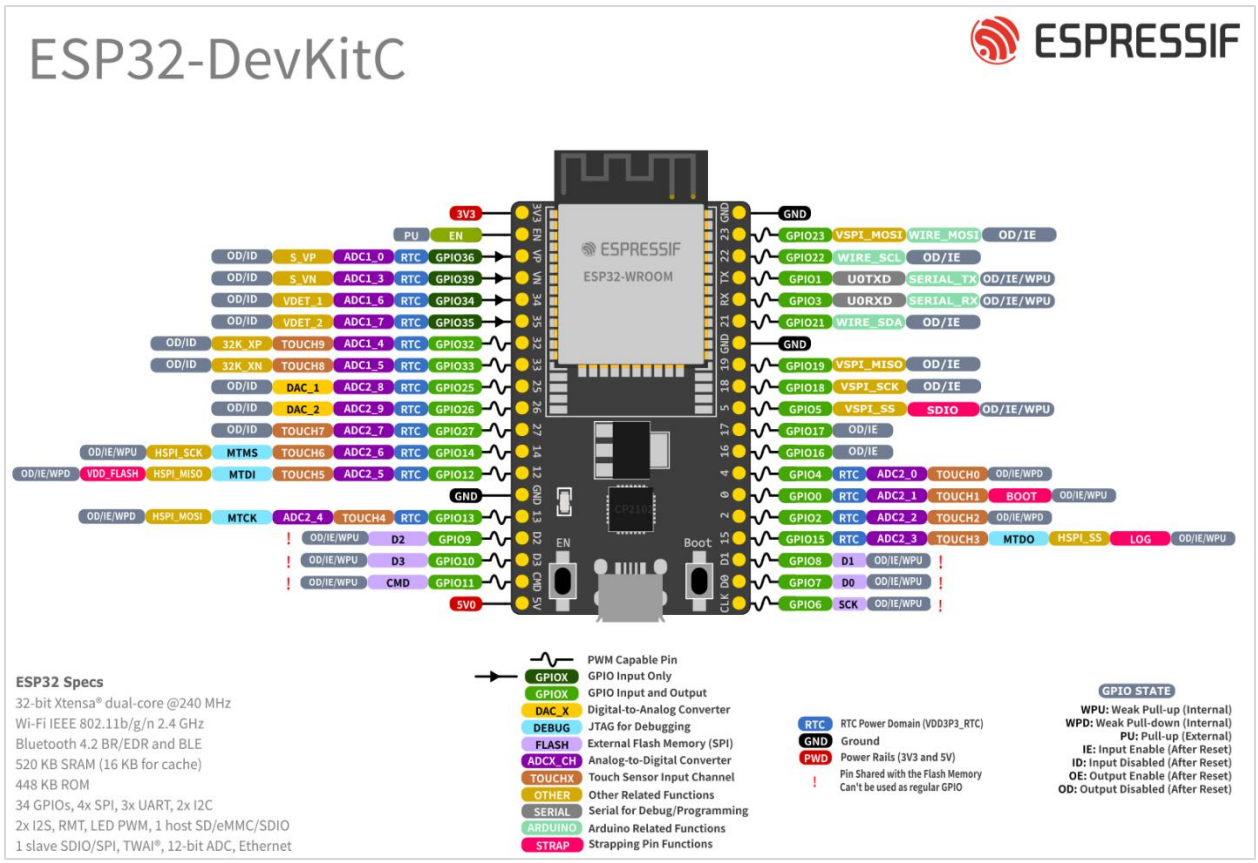
Getting Started ESP32 Programming

The ESP32 is a popular, low-cost System-on-Chip (SoC) microcontroller with integrated WiFi and Bluetooth, designed by Espressif Systems for IoT (Internet of Things) and embedded applications, featuring dual-core (or single-core) processors, low power consumption, and versatile peripherals, making it ideal for smart home devices, wearables, and complex connectivity projects, programmable via C/IDF Library, MicroPython, and other platforms. In this lab, we use the ESP32-DevKitC V4, a small-sized ESP32-WROOM development board. Most of the I/O pins are broken out to the pin headers on both sides for easy interfacing.

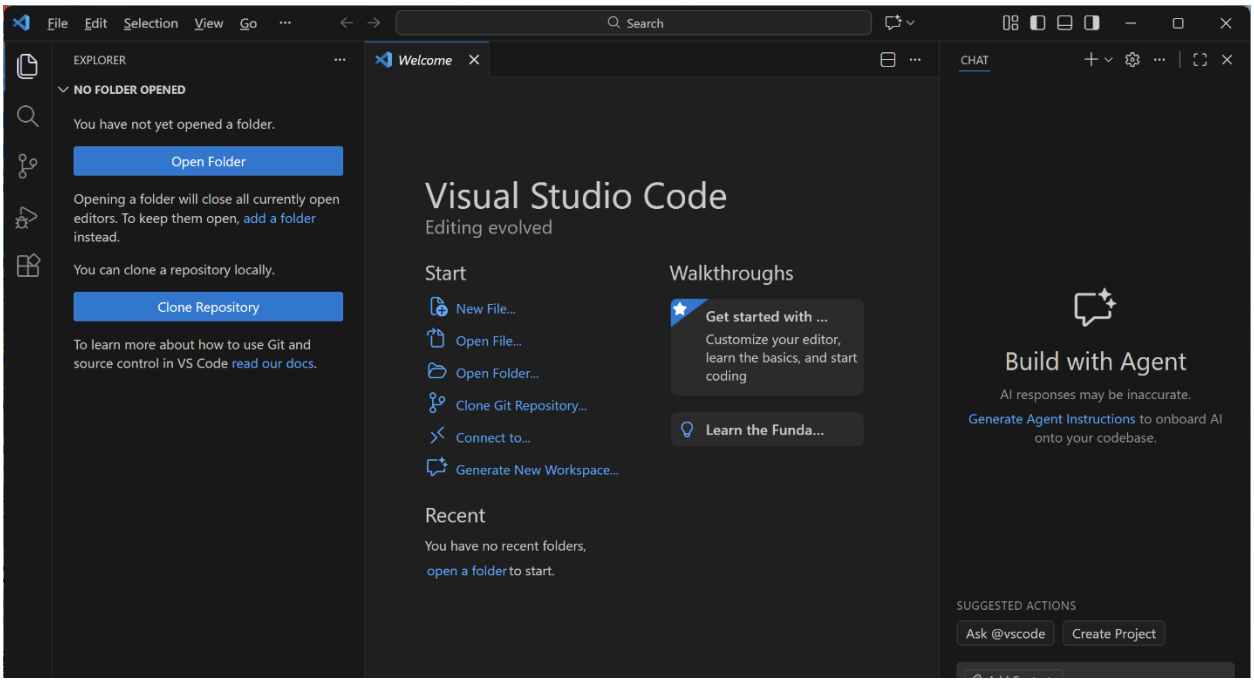
c



In this document, we aim to explain how to start programming the ESP32 using Visual Studio Code (VS Code) and the ESP-IDF (Espressif IoT Development Framework) in the C language. The following steps outline the process of setting up the programming environment, creating, building, and running the first project on the ESP32 development board.

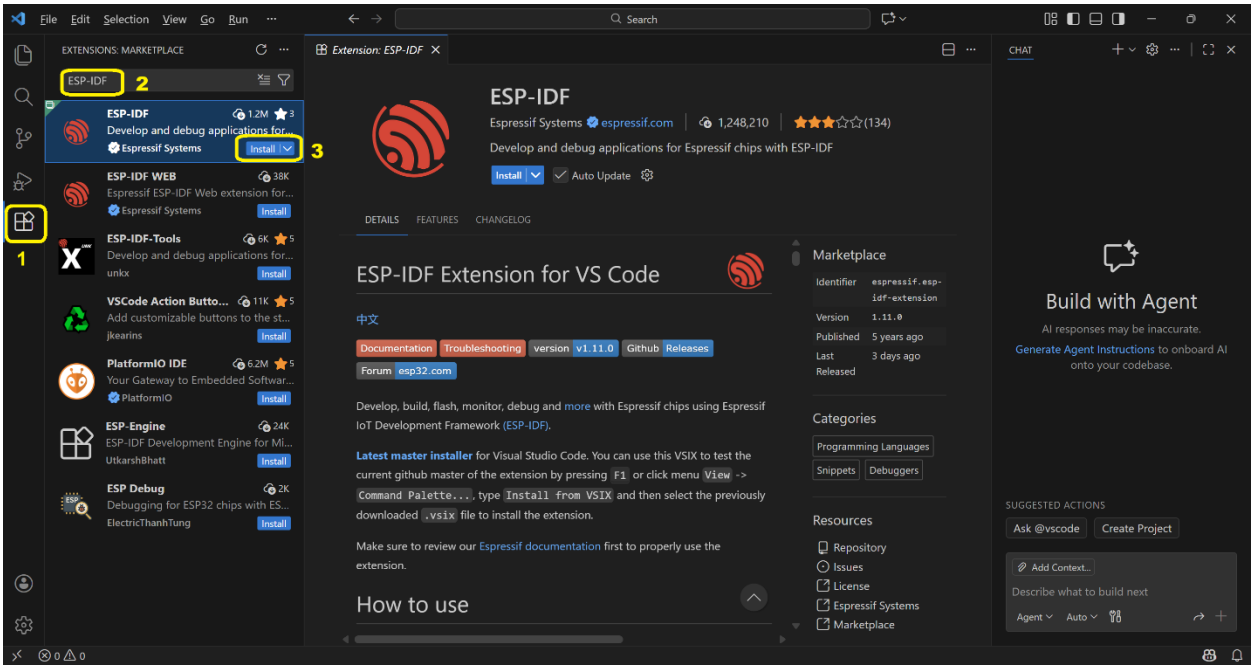
Step 1: Installing VSCode

Download and install the latest version of VSCode.



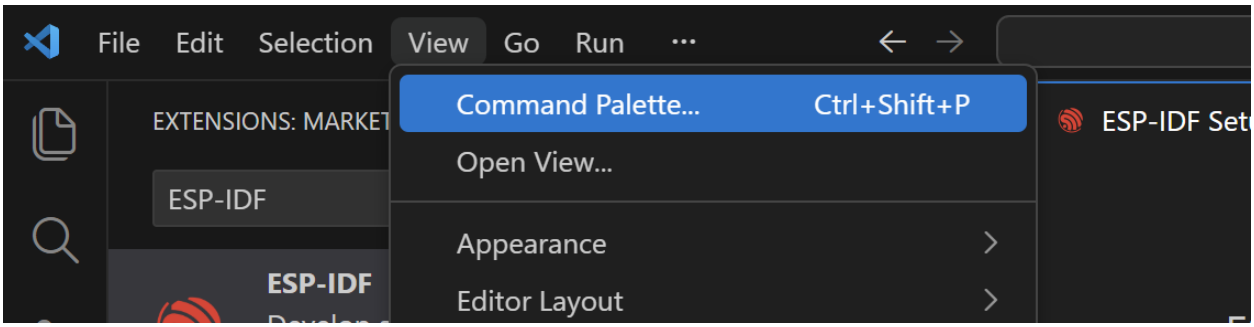
Step 2: Installing Extensions

In the Extension section, search for "ESP-IDF" and install this extension as illustrated in the figure.

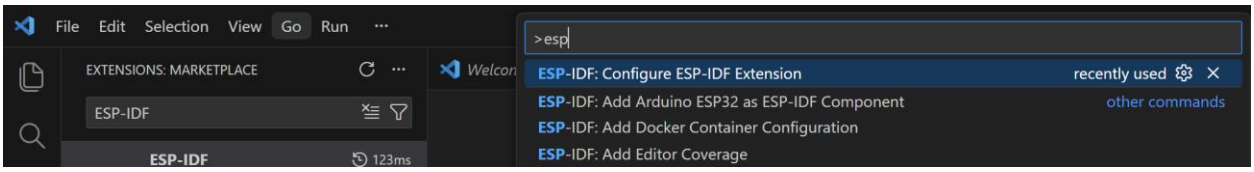


Step 3: Downloading and Configuring ESP-IDF

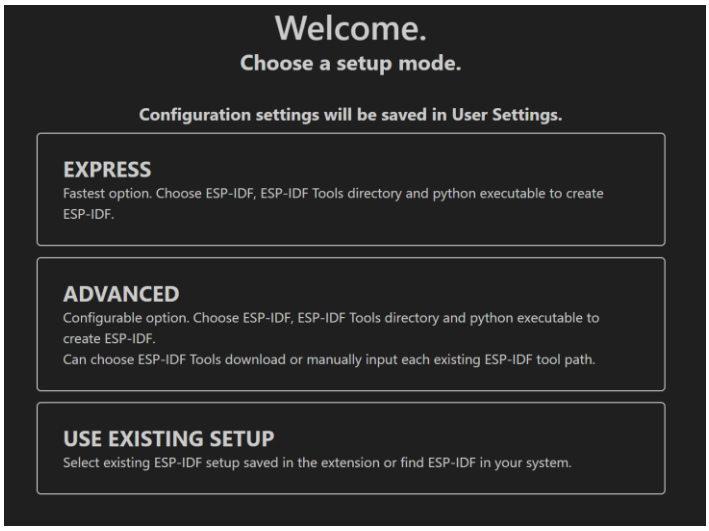
Open View → Command Palette. You can access all the IDF commands through this menu.



Then, type IDF-ESP: Configure ESP-IDF Extension and select it.



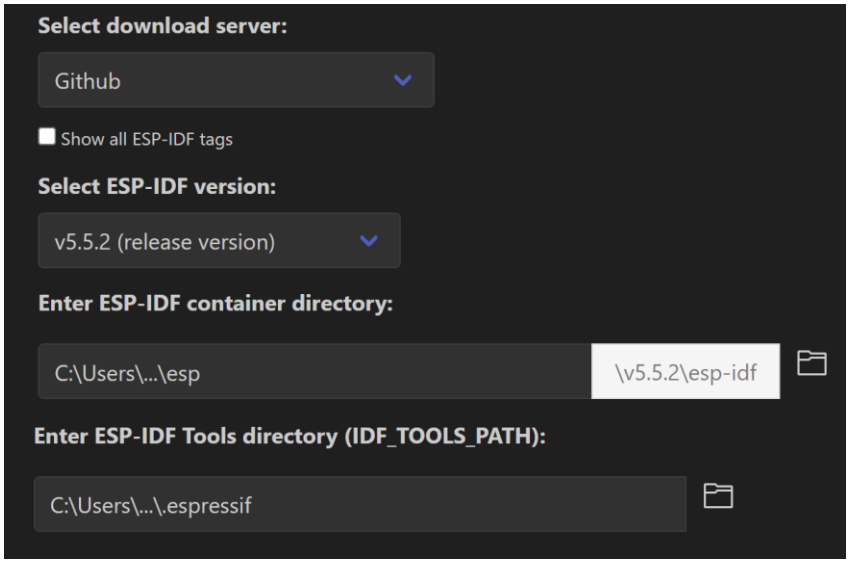
In the opened window, select the "Express" option.



You can see the options for installing ESP-IDF libraries and tools.

- ESP-IDF Version: Select the latest released version.
- ESP-IDF Container: Enter IDF path.
- ESP-IDF Tools Directory: Enter ESP-IDF tools path.

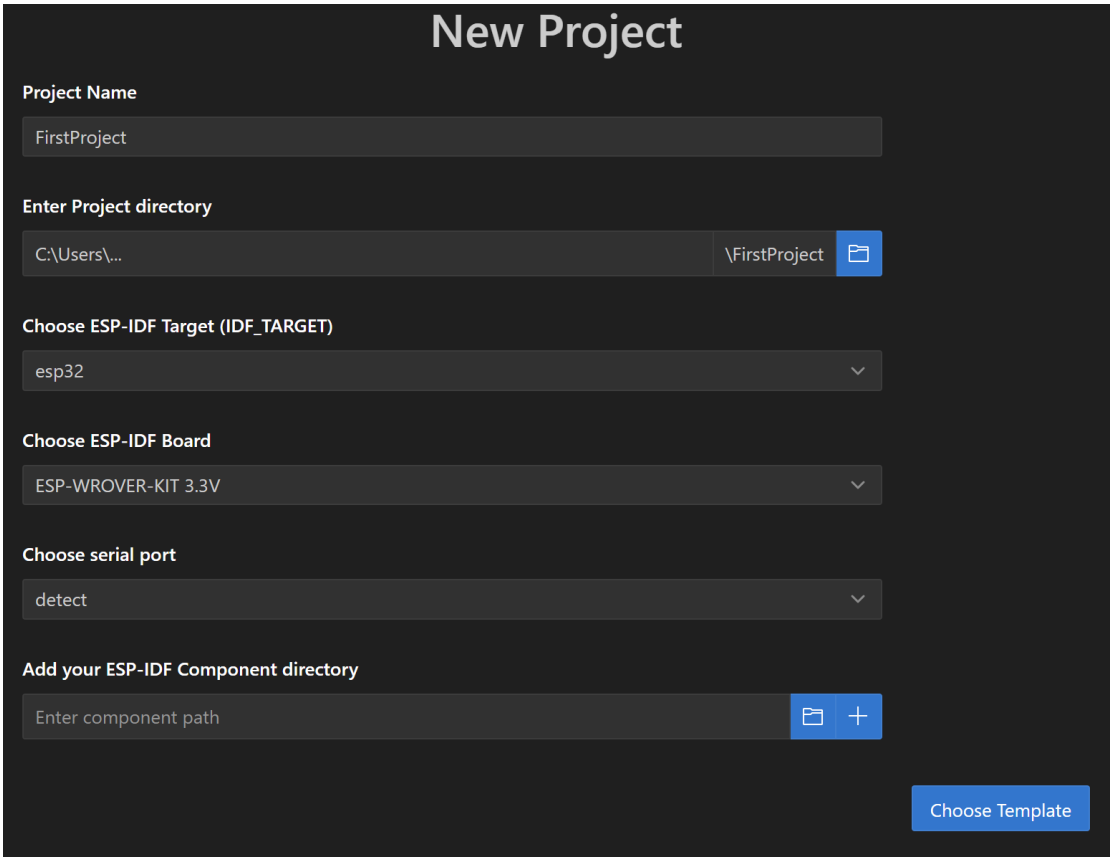
Ensure that IDF_TOOLS_PATH does not contain spaces to prevent build issues. Also, ensure that IDF_TOOLS_PATH is not the same directory as IDF_PATH.



Click Install and wait for the tools to be installed.

Step 4: Creating a New Project

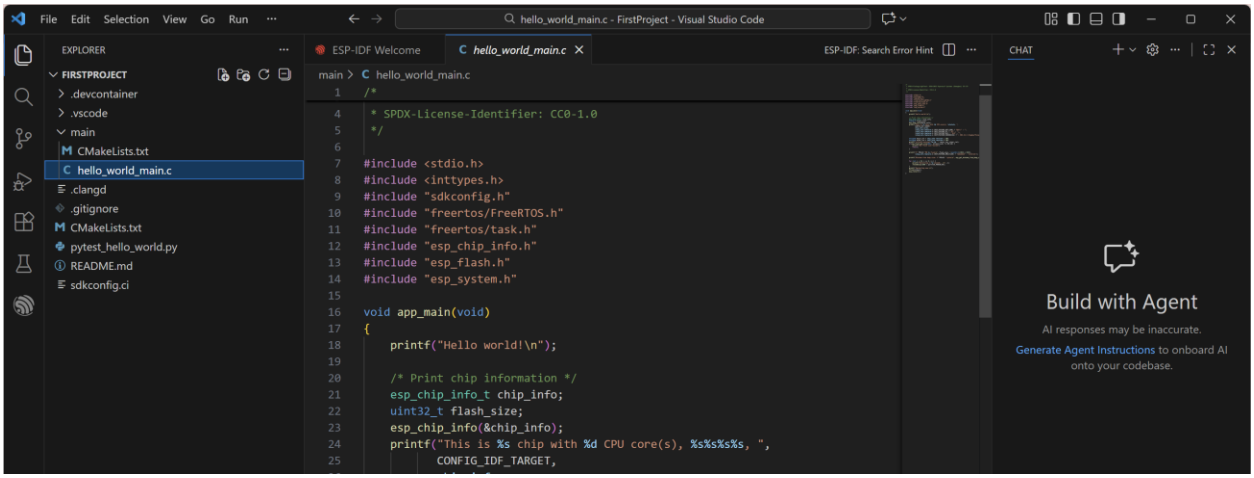
From Command Palette, select "ESP-IDF: New Project". You can select the name and path of the new project. The Target and Board fields should be left on ESP32 and ESP-WROVER-LIT 3.3V, respectively.



The serial port will be configured later. Then, click on the Choose Template button and pick the hello_world example from the path below:

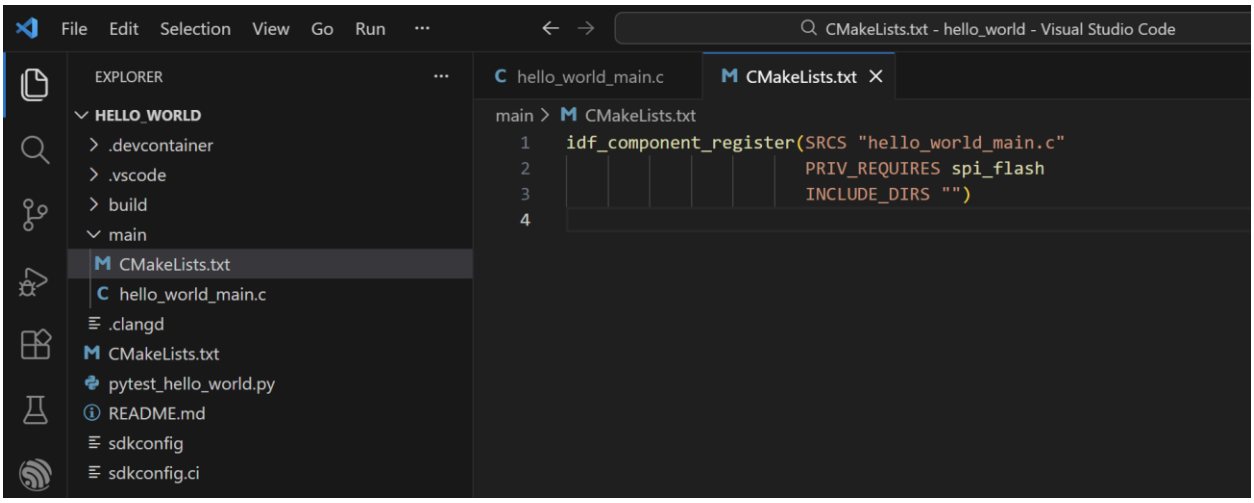
ESP-IDF → get-started → hello_world

Now, click on "Create project using template hello_world" and open the created project.



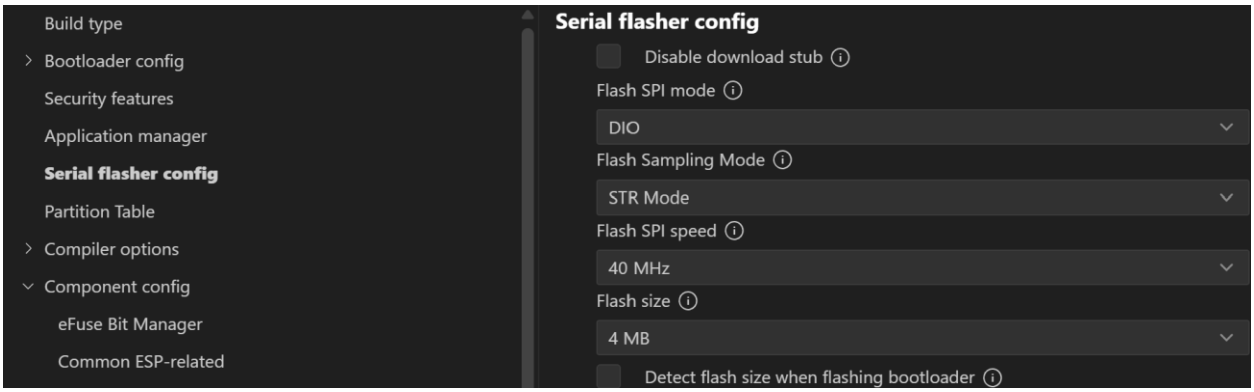
In the Explorer section on the left side, the project's folders and files are presented. Two important files are located in the "main" folder. The file "hello_world_main.c" contains the main function of the code. In this code, the chip information and other strings are printed through the UART.

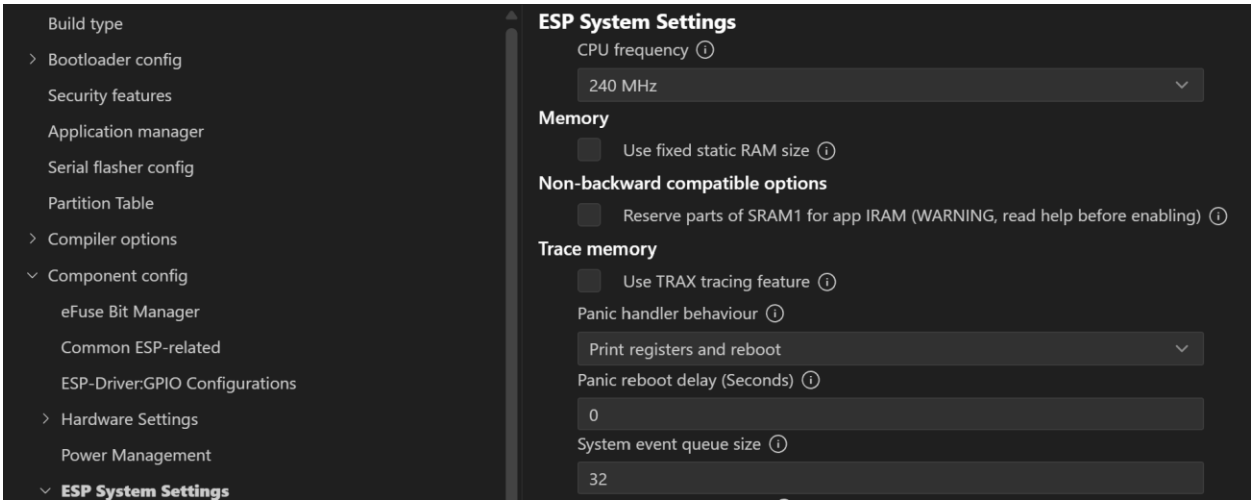
The other file, "CMakeLists.txt", determines the source files, libraries, and modules required for the project. The source files (.c files) can be named after the SRCS keyword, which is hello_world_main.c in this project. Additionally, we should provide the modules required in the code so that the system can create the related libraries. For example, if we want to use WiFi, we should add "esp_wifi" after the PRIV_REQUIRES keyword. As this project needs SPI Flash memory information, the "spi_flash" module has been added.



Step 5: Hardware Configuration

From Command Palette, run "ESP-IDF: SDK Configuration Editor (Menuconfig)". Several hardware configurations (esp32) are available in the opened window. We know that the module uses 4 MB of flash memory. Therefore, from the "Serial flash config" section, we can change the flash size to 4 MB. Furthermore, the system operates at 160 MHz by default. You can change the CPU frequency to 240 MHz in the "ESP System Settings" section of "Component config".





Please note that modifying certain parameters in this section may lead to irreversible changes and could potentially result in permanent damage to the device.

Step 6: Build The Project

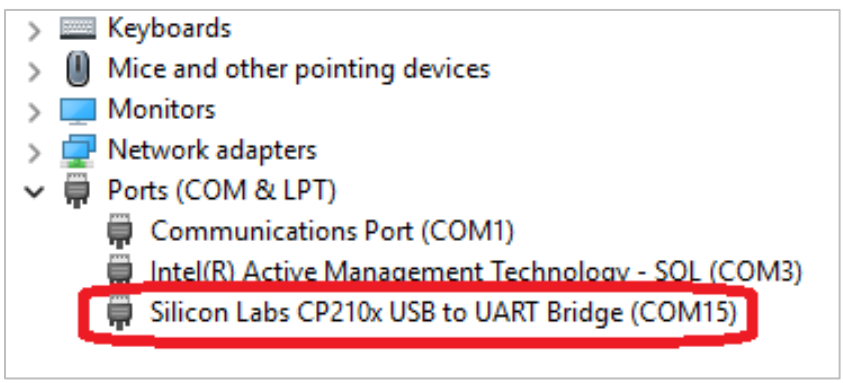
Compiling the whole project can be done through "ESP-IDF: Build Your Project". If the project's building is successful, the details of the bin file should be demonstrated.

In this step, the libraries written in the "main/CMakelists.txt" file will be created. Additionally, you can view the output logs (Output).

```
                          Memory Type Usage Summary
+-------------------------------------------------------------------------------+
| Memory Type/Section   | Used [bytes] | Used [%] | Remain [bytes] | Total [bytes] |
|-----------------------+--------------+----------+----------------+---------------|
| IRAM                  |        52039 |     39.7 |          79033 |        131072 |
|    .text              |        51011 |    38.92 |                |               |
|    .vectors           |         1028 |     0.78 |                |               |
| Flash Code            |        46902 |          |                |               |
|    .text              |        46902 |          |                |               |
| Flash Data            |        29524 |          |                |               |
|    .rodata            |        29268 |          |                |               |
|    .appdesc           |          256 |          |                |               |
| DRAM                  |        10832 |     5.99 |         169904 |        180736 |
|    .data              |         8688 |     4.81 |                |               |
|    .bss               |         2144 |     1.19 |                |               |
| RTC SLOW              |           56 |     0.68 |           8136 |          8192 |
|    .force_slow        |           32 |     0.39 |                |               |
|    .rtc_slow_reserved |           24 |     0.29 |                |               |
+-------------------------------------------------------------------------------+
Total image size: 137185 bytes (.bin may be padded larger)
```

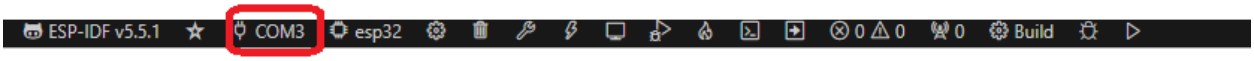
Step 7: Flashing ESP32 chip

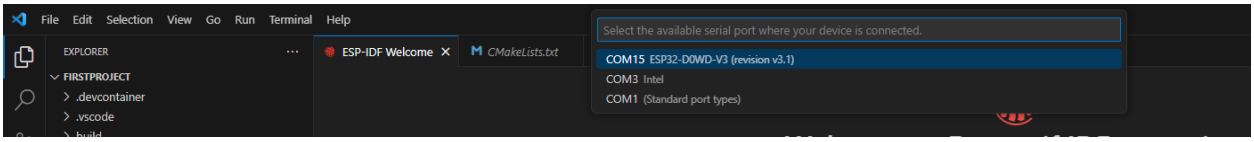
We must install the Virtual COM Port (VCP) driver for writing the binary file (bin) through VCP. The installation file has been provided. After installing this file, connect ESP32 board through USB and open Device Manager.



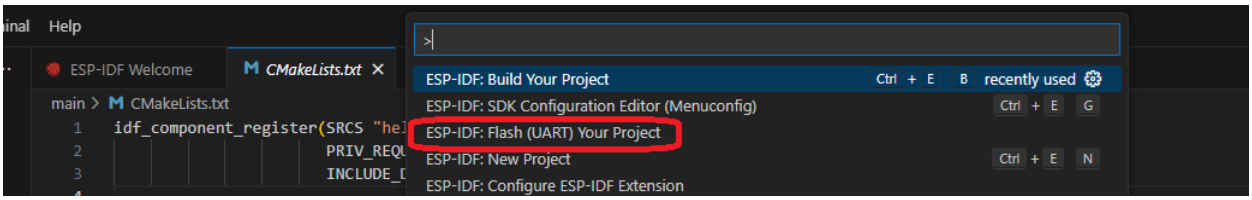
The COM port associated with the ESP32 can be identified in the Windows Device Manager under the Ports (COM & LPT) section. The assigned COM port number may vary between computers. In the system used for this demonstration, the ESP32 serial interface is assigned to COM15.

Return to VSCode and click on COM port at **the bottom of the window** and select the associated COM port at **the top menu**.

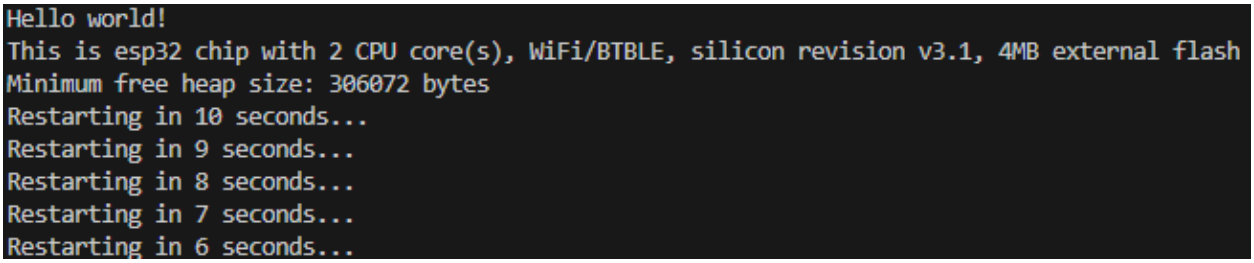




Finally, from Command Palette, select "ESP-IDF: Flash (UART) Your Project".



After finishing, the output of the serial port can be seen in the terminal. To open the terminal, you can click on the Monitor Device button at the bottom of the window.



Troubleshooting:

In some cases, the ESP development tools may not be installed correctly, resulting in errors during the installation process. In addition, the installation procedure may be interrupted due to network connectivity issues or system failures. Under such circumstances, it is necessary to remove the previously installed files and repeat the installation. To do so, follow the steps below:

1. Open Command Palette (press shortcut F1) and type **ESP-IDF: Clear Saved ESP-IDF Setups**. Select the command to remove all ESP-IDF settings.
2. Navigate to View → Extensions or use the keyboard shortcut Ctrl+Shift+X in Windows/Linux.
3. Search for ESP-IDF and click the Uninstall button.
4. Go to below address and delete the ESP-IDF plugin folder:

```
%USERPROFILE%/.vscode/extensions/
```

5. Delete ESP tools folders. If you did not change installation paths when installing, you can find them at the following addresses:

```
C:\Users\<your username>\.espressif

C:\Users\<your username>\esp
```

Before installing ESP extensions and tools, you can also uninstall all versions of Python on your computer and install Python 3.10.x, which is the most compatible version.

Using an Existing Project:

If you import a project from a computer that the project has not been created, you may face some build problems. The easiest way to use an existing project in this case is to create a new project (Hello World) and copy the contect of main folder of the existing project into the main folder of the new project. Then, build.